

Задание 19 — Кросс компиляция

1) загружаем и устанавливаем исходный код zlib версии 1.3.2.



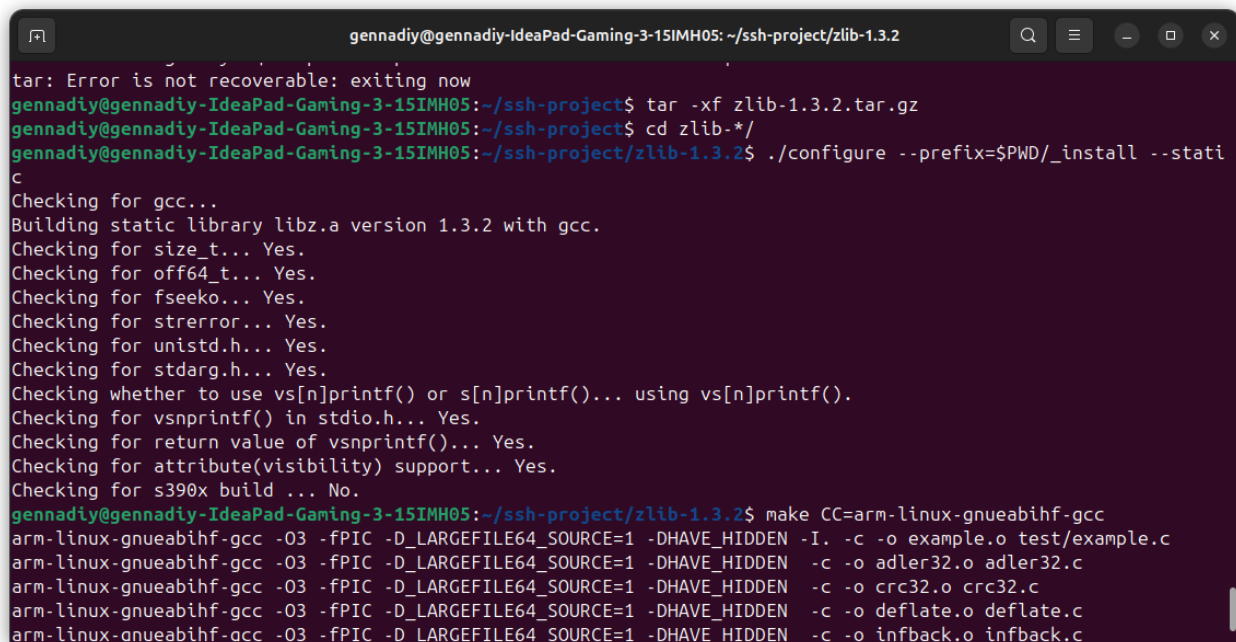
```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/zlib-1.3.2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ..
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ mkdir ~/ssh-project
cd ~/ssh-project
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ mkdir target
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ wget https://zlib.net/zlib-1.3.2.tar.gz
--2026-06-15 15:33:09-- https://zlib.net/zlib-1.3.2.tar.gz
Распознаётся zlib.net (zlib.net)... 85.187.148.2
Подключение к zlib.net (zlib.net)|85.187.148.2|:443... соединение установлено.
HTTP-запрос отправлен. Ожидание ответа... 200 OK
Длина: 1502830 (1,4M) [application/x-gzip]
Сохранение в: 'zlib-1.3.2.tar.gz'

zlib-1.3.2.tar.gz      100%[=====>] 1,43M  753KB/s   за 1,9s
2026-06-15 15:33:13 (753 KB/s) - 'zlib-1.3.2.tar.gz' сохранён [1502830/1502830]

gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ tar -xvf zlib.tar.gz
tar: zlib.tar.gz: Функция open завершилась с ошибкой: Нет такого файла или каталога
tar: Error is not recoverable: exiting now
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ tar -xvf zlib-1.3.2.tar.gz
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ cd zlib-*/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$ ./configure --prefix=$PWD/_install --static
Checking for gcc...
```

Рисунок 1 — установка zlib

2) конфигурацию сборки выполняем командой `./configure --prefix=$PWD/_install --static`



```
tar: Error is not recoverable: exiting now
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ tar -xvf zlib-1.3.2.tar.gz
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project$ cd zlib-*/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$ ./configure --prefix=$PWD/_install --static
Checking for gcc...
Building static library libz.a version 1.3.2 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for strerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
Checking for s390x build ... No.
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$ make CC=arm-linux-gnueabi-gcc
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inffback.o inffback.c
```

Рисунок 2 — конфигурация сборки

3) запускаем сборку (компиляцию) командой `make CC=arm-linux-gnueabi-gcc` и затем выполняем `make install` для локальной установки

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/zlib-1.3.2
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o minigzip64 minigzip64.o -L. libz.a
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$ make install
rm -f /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib/libz.a
cp libz.a /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib
chmod 644 /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib/libz.a
rm -f /home/gennadiy/ssh-project/zlib-1.3.2/_install/share/man/man3/zlib.3
cp zlib.3 /home/gennadiy/ssh-project/zlib-1.3.2/_install/share/man/man3
chmod 644 /home/gennadiy/ssh-project/zlib-1.3.2/_install/share/man/man3/zlib.3
rm -f /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
cp zlib.pc /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib/pkgconfig
chmod 644 /home/gennadiy/ssh-project/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
rm -f /home/gennadiy/ssh-project/zlib-1.3.2/_install/include/zlib.h /home/gennadiy/ssh-project/zlib-1.3.2/_install/include/zconf.h
cp zlib.h zconf.h /home/gennadiy/ssh-project/zlib-1.3.2/_install/include
chmod 644 /home/gennadiy/ssh-project/zlib-1.3.2/_install/include/zlib.h /home/gennadiy/ssh-project/zlib-1.3.2/_install/include/zconf.h
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$ ls
adler32.c      deflate.c      gzlib.o       inflate.o     MODULE.bazel  win32
adler32.o     deflate.h      gzread.c      infrees.c    msdos         zconf.h
amiga         deflate.o      gzread.o      infrees.h    os400         zconf.h.in
BUILD.bazel   doc           gzwrite.c     infrees.o    qnx           zlib.3
ChangeLog     example       gzwrite.o     _install     README        zlib.3.pdf
CMakelists.txt example64.o    INDEX        libz.a       README-cmake.md zlibConfig.cmake.in
compress.c    example64.o   infback.c     LICENSE      test          zlib.h
compress.o    example.o     infback.o     Makefile     treebuild.xml zlib.map
configure     examples      inffast.c     Makefile.in  trees.c       zlib.pc
configure.log FAQ           inffast.h     make_vms.com trees.h       zlib.pc.cmakein
contrib       gzclos.c     inffast.o     minigzip     trees.o       zlib.pc.in
crc32.c       gzclos.o     inffix.h      minigzip64  uncompr.c    zutil.c
crc32.h       gzguts.h     inflate.c     minigzip64.o uncompr.o    zutil.h
crc32.o       gzlib.c      inflate.h     minigzip.o   watcom       zutil.o
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/zlib-1.3.2$
```

Рисунок 3 — завершение установки zlib

4) клонируем исходный код OpenSSL 3.0.21 командой `git clone --branch openssl-3.0.21 --depth 1 https://github.com/openssl/openssl.git openssl-3.0.21`.

Далее выполняем конфигурацию сборки OpenSSL командой `./Configure linux-armv4 zlib --with-zlib-include="$PWD/../zlib-1.3.2/_install/include"`

`--with-zlib-lib="$PWD/../zlib-1.3.2/_install/lib" --prefix="$PWD/_install" --cross-compile-prefix=arm-linux-gnueabi-`

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/openssl-3.0.21
rm -f Makefile
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/openssl-3.0.21$ ./Configure linux-armv4 zlib --with-zlib-include="$PWD/../zlib-1.3.2/_install/include" --with-zlib-lib="$PWD/../zlib-1.3.2/_install/lib" --prefix="$PWD/_install" --cross-compile-prefix=arm-linux-gnueabihf-
Configuring OpenSSL version 3.0.21 for target linux-armv4
Using os-specific seed configuration
Created configdata.pm
Running configdata.pm
Created Makefile.in
Created Makefile
Created include/openssl/configuration.h

*****
***
***   OpenSSL has been successfully configured   ***
***                                           ***
***   If you encounter a problem while building, please open an ***
***   issue on GitHub <https://github.com/openssl/openssl/issues> ***
***   and include the output from the following command: ***
***                                           ***
***       perl configdata.pm --dump ***
***                                           ***
***   (If you are new to OpenSSL, you might want to consult the ***
***   'Troubleshooting' section in the INSTALL.md file first) ***
***                                           ***
*****
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/openssl-3.0.21$
```

Рисунок 4 — конфигурация сборки

5) выполняем компиляцию (make -j8) и локальную установку (make install_sw)

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/openssl-3.0.21
rm -f test/uitest
$LDCCMD:-arm-linux-gnueabi-gcc} -pthread -Wa,--noexecstack -Wall -O3 -L. -L/home/gennadiy/ssh-project/openssl-3.0.21/../zlib-1.3.2/_install/lib \
-o test/uitest \
  apps/lib/uitest-bin-apps_ui.o test/uitest-bin-uitest.o \
  -lssl test/libtestutil.a -lcrypto -lz -ldl -pthread -latomic
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/openssl-3.0.21$ make install_sw
"make" depend && "make" _build_libs
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
*** Installing runtime libraries
install libcrypto.so.3 -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/libcrypto.so.3
install libssl.so.3 -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/libssl.so.3
*** Installing development files
install ./include/openssl/aes.h -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/include/openssl/aes.h
install ./include/openssl/asn1.h -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/include/openssl/asn1.h
install ./include/openssl/asn1_mac.h -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/include/openssl
```

Рисунок 5 — выполнение установки в директорию

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/openssl-3.0.21
install libcrypto.pc -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/pkgconfig/libcrypto.pc
install libssl.pc -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/pkgconfig/libssl.pc
install openssl.pc -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/pkgconfig/openssl.pc
"make" depend && "make" _build_modules
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Nothing to be done for '_build_modules'.
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
*** Installing engines
install engines/capi.so -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/engines-3/capi.so
install engines/loader_attic.so -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/engines-3/loader_attic.so
install engines/padlock.so -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/engines-3/padlock.so
*** Installing modules
install providers/legacy.so -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/lib/openssl-modules/legacy.so
"make" depend && "make" _build_programs
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Entering directory '/home/gennadiy/ssh-project/openssl-3.0.21'
make[1]: Nothing to be done for '_build_programs'.
make[1]: Leaving directory '/home/gennadiy/ssh-project/openssl-3.0.21'
*** Installing runtime programs
install apps/openssl -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/bin/openssl
install tools/c_rehash -> /home/gennadiy/ssh-project/openssl-3.0.21/_install/bin/c_rehash
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/openssl-3.0.21$
```

Рисунок 6 — завершение сборки и установки библиотек OpenSSL

6) клонируем исходный код OpenSSH командой `git clone --branch V_9_8_P1 --depth 1 https://github.com/openssh/openssh-portable.git openssh-9.8p1`. Далее выполняем конфигурацию сборки OpenSSH командой

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/ssh-project/openssh-9.8p1
checking host system type... Invalid configuration 'arm-linux-gnueabihr-': OS not recognized
configure: error: /bin/bash ./config.sub arm-linux-gnueabihr- failed
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/ssh-project/openssh-9.8p1$ ./configure --host=arm-linux-gnueabihr --with-zlib="$PWD/../zlib-1.3.2/_install" --with-ssl-dir="$PWD/../openssl-3.0.21/_install" --prefix="$PWD/_install"
checking for arm-linux-gnueabihr-cc... no
checking for arm-linux-gnueabihr-gcc... arm-linux-gnueabihr-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether arm-linux-gnueabihr-gcc accepts -g... yes
checking for arm-linux-gnueabihr-gcc option to enable C11 features... none needed
checking if arm-linux-gnueabihr-gcc supports C99-style variadic macros... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... arm-unknown-linux-gnueabihr
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes
checking for sys/stat.h... yes
checking for sys/types.h... yes
checking forunistd.h... yes
```

Рисунок 7 — конфигурация сборки

7) выполняем компиляцию `make -j8`

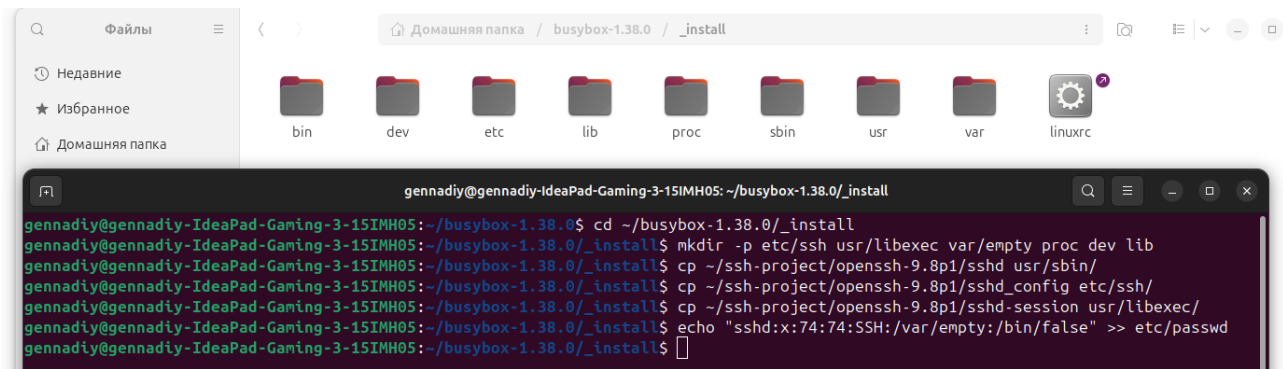


Рисунок 10 — копируем исполняемый файлы и конфигурацию OpenSSH в корневую файловую систему

9) переносим динамические библиотеки в нашу системную директорию lib

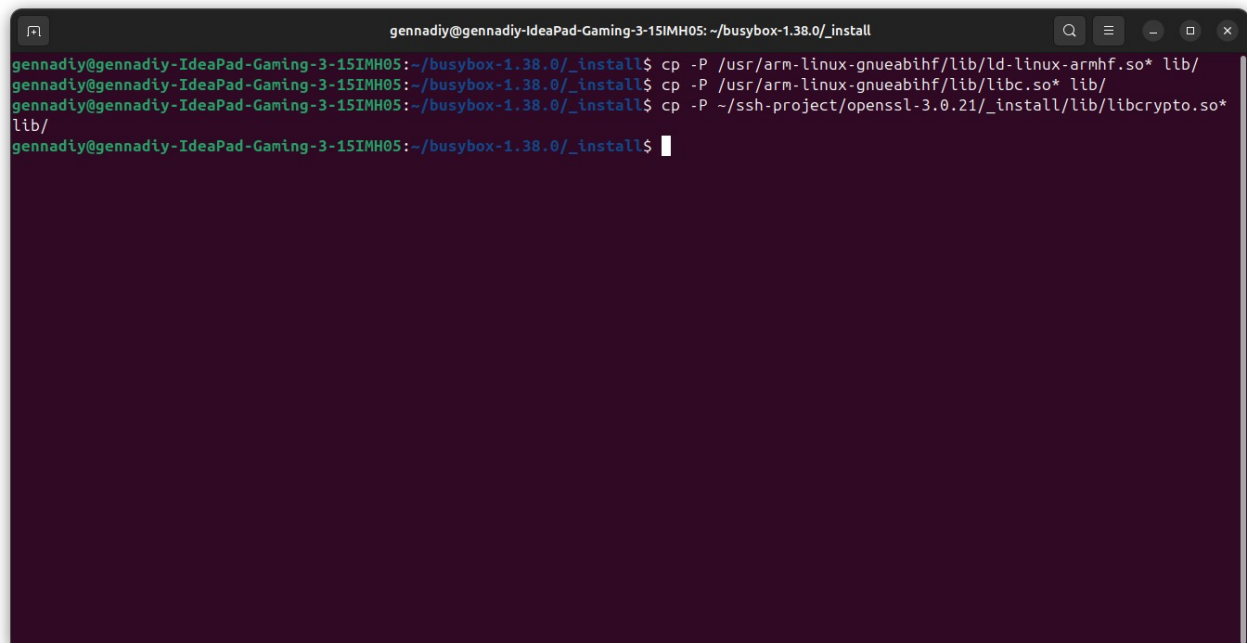


Рисунок 11 — перенос динамических библиотек в lib

10) генерация ключа шифрования хоста утилитой ssh-keygen для запуска сервера

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ~/busybox-1.38.0/_install
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ ssh-keygen -t rsa -f etc/ssh/ssh_host_
rsa_key -N ""
Generating public/private rsa key pair.
Your identification has been saved in etc/ssh/ssh_host_rsa_key
Your public key has been saved in etc/ssh/ssh_host_rsa_key.pub
The key fingerprint is:
SHA256:tQNKtbfXz5JmuUUMd09R12IzgBV3tnlqm6y5tDd/1hQ gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05
The key's randomart image is:
+---[RSA 3072]-----+
|      oo.o=oo.>%|
|      o  o. o=OB|
|      . . . .**|
|      o . . Eo|
|      S o   O =|
|      . = *.|
|      . =..|
|      . +o +|
|      =o +o|
+-----[SHA256]-----+
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$
```

Рисунок 12 — генерация ключа шифрования

12) Далее вручную записываем конфигурационные файлы passwd и shadow. Пароль для входа под root отсутствует.

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/bb-project$ cd ~/busybox-1.38.0/_install
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ echo "root:x:0:0:root:/root:/bin/ash"
>> etc/passwd
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$ echo "root::20305:0:99999:7:::" > etc/
shadow
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/busybox-1.38.0/_install$
```

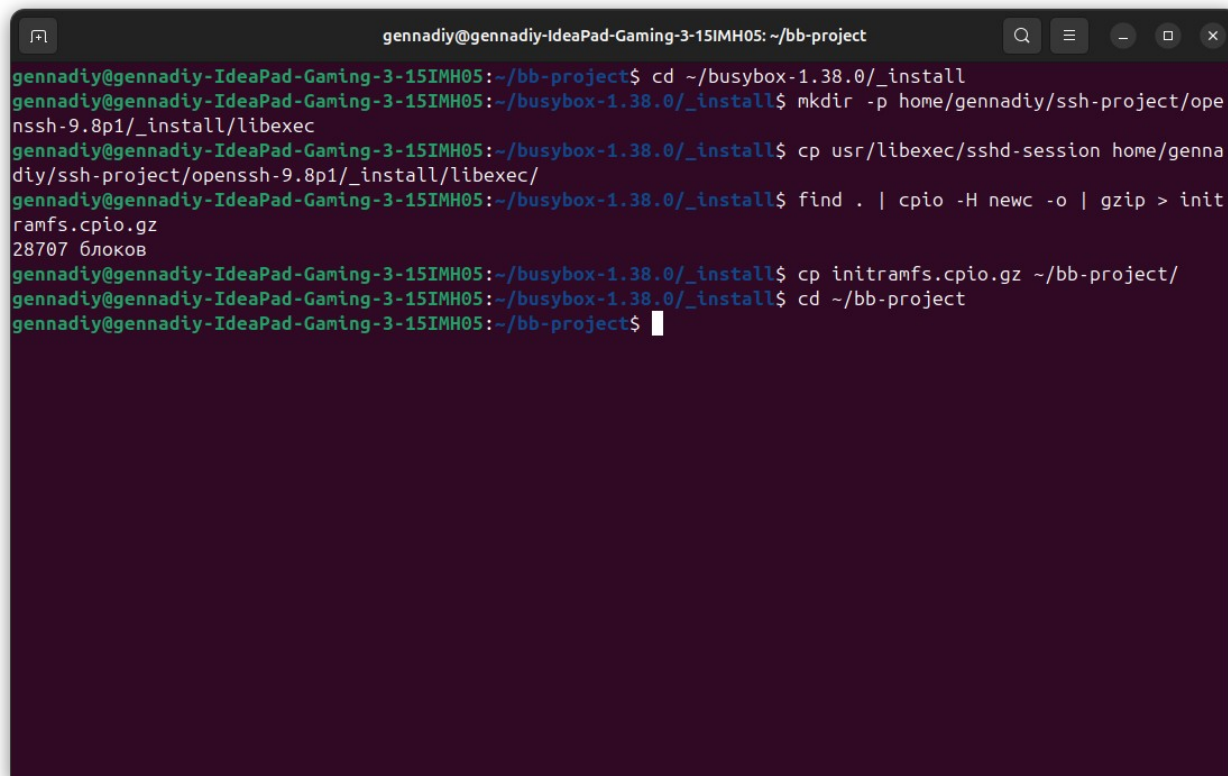
Рисунок 13 — конфигурация системы пользователей

13) Обновленную структуру файловой системы архивируем и копируем в рабочую директорию bb-project.

В ходе тестирования было обнаружено, что исходный код OpenSSH скомпилирован с префиксом путей хоста (\$PWD/_install). Для обеспечения работоспособности была создана необходимая директория и в нее скопирован модуль sshd-session:

```
mkdir -p home/gennadiy/ssh-project/openssh-9.8p1/_install/libexec
cp usr/libexec/sshd-session home/gennadiy/ssh-project/openssh-9.8p1/_install/libexec/
```

Запускаем систему в эмуляторе командой QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -initrd initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/bin/ash" -nographic

A screenshot of a terminal window with a dark background and light-colored text. The window title is "gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project". The terminal shows a series of commands and their outputs. The user navigates to ~/busybox-1.38.0/_install, creates a directory home/gennadiy/ssh-project/openssh-9.8p1/_install/libexec, copies the file usr/libexec/sshd-session to that directory, and then uses the find command to create a gzipped archive of the current directory's contents. The output of find shows 28707 blocks. Finally, the user copies the archive to ~/bb-project/initramfs.cpio.gz and returns to the ~/bb-project directory.

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project$ cd ~/busybox-1.38.0/_install
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install$ mkdir -p home/gennadiy/ssh-project/ope
nssh-9.8p1/_install/libexec
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install$ cp usr/libexec/sshd-session home/genna
diy/ssh-project/openssh-9.8p1/_install/libexec/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install$ find . | cpio -H newc -o | gzip > init
ramfs.cpio.gz
28707 блоков
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install$ cp initramfs.cpio.gz ~/bb-project/
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/busybox-1.38.0/_install$ cd ~/bb-project
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project$
```

Рисунок 14 — перенос обновленной файловой системы в рабочую директорию

14) Внутри QEMU выполняем ряд команд:

mount -t devtmpfs devtmpfs /dev — монтируем вирт. файловую систему устройств ядра в dev для получения dev/null

mount -t proc proc /proc — монтируем вирт. файловую систему процессов

chown 0:0 /var/empty — смена UID:GID на root с 0:0

chmod 700 /var/empty — ограничение прав доступа для var/empty, даем доступ только root

/usr/sbin/sshd -f /etc/ssh/sshd_config -h /etc/ssh/ssh_host_rsa_key — запуск сервера с переопределением пути к конфигурации и передачей пути к ключе хоста.


```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/bb-project
[ 2.647925] clk: Disabling unused clocks
[ 2.648536] PM: genpd: Disabling unused power domains
[ 2.710072] Freeing unused kernel image (initmem) memory: 2048K
[ 2.717667] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ # [ 2.755866] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:mot
herboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kni/serio1/input/inpu
t2
[ 13.274697] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 13.274990] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 13.275062] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 13.275110] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

~ # mount -t devtmpfs devtmpfs /dev
~ # mount -t proc proc /proc
~ # chown 0:0 /var/empty
~ # chmod 700 /var/empty
~ # /usr/sbin/sshd -f /etc/ssh/sshd_config -h /etc/ssh/ssh_host_rsa_key

[ 33.751204] random: crng init done
~ #
~ # netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 :::22                  :::*                    LISTEN
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags               Type                   State                  I-Node Path
~ #
```

Рисунок 15 — результат выполнения